

frame t − 1

Fused maps Z
strides 8, 16, 32

Z, ρ

S_{t-2}

Temporal
State Memory

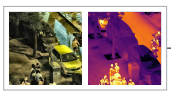
Neck and
one-to-one head

Detections

ByteTrack
tracks

causal: frames in order, no future frame used

state S_{t-1} warped by H_t;
step scaled by τ_t/τ_0 and ρ_t



frame t

Fused maps Z
strides 8, 16, 32

Z, ρ

Temporal
State Memory

Neck and
one-to-one head

Detections

ByteTrack
tracks

S_t to frame t+1; reset at a scene cut

Evaluation on KAIST

Two modes
same weights, memory off or on at test time

Conditions
clean, thermal lost (flagged or not), visible darkened, thermal noise

Detection
log-average miss rate, all frames and failed frames

Tracking
MOTA, IDF1, identity switches on four test sequences